Mathematics III_2025_PYQ

Group-A (Very Short Answer Type Questions)

Answer any ten of the following:

1. Find f_x and f_y where $f(x, y) = x^2 + xy$.
2. The value of $\int_1^0 \int_0^1 (x + y) dx\, dy$ is _____.
3. The general solution of $px + \frac{a}{p}$, where $p = \frac{dy}{dx}$ is _____.
4. The value of $\frac{1}{D}(x^2)$ is _____.
5. If a path is considered as a subgraph, then the degree of the intermediate vertices is _____.

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6. The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if _____.
7. If $u + v = x, uv = y$, then $\frac{\partial(x,y)}{\partial(u,v)} =$_____.
8. The value of $\int_0^{\frac{\pi}{2}} \int_0^1 r \sin\theta\, dr\, d\theta$ is _____.
9. The singular solution of the equation $y = px + a\sqrt{1 + p^2}$ is _____.
10. The value of $D(\sin^3 x)$ is _____.
11. A binary tree has exactly _____.
12. The coefficient of x^3 in the infinite series expansion of $f(x) = \sin x$ about $x = \frac{\pi}{2}$ is _____.

Group-B (Short Answer Type Questions)

Answer any three of the following:

1. Show that $\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u=\frac{-9}{(x+y+z)^2}$ if $u=\log(x^3+y^3+z^3-3xyz)$.
2. Evaluate $\iint_R xy(x+y)\,,dx,dy$ over the region R bounded by $y=x^2$and $y\ =\ x$.
3. Solve the ODE: $xp^2+(y-x)p-y=0$, where $p=\frac{dy}{dx}$.
4. Find the Taylor's series expansion of $\ln x$.
5. Solve by variation of parameters, $\frac{d^2y}{dx^2}+a^2y=\cos(ax)$.

Group-C (Long Answer Type Questions)

Answer any three of the following:

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1. (a) Find the maxima and minima of the function $f(x,y)=x^3+y^3-3x-12y+20$. Also find the saddle points.

   (b) Find $div\,\vec{F}$ and $curl\,\vec{F}$ where $\vec{F}=\nabla(x^3+y^3+z^3-3xyz)$.
2. (a) Evaluate $\iint y\,dx\,dy$ over the part of the plane bounded by $y\ =\ x$ and the parabola

   $y=4x-x^2$.

   (b) Change the order of integration and hence evaluate the integral $\int_0^1\int_{e^x}^{e}\frac{dx,dy}{y^2\log y}$.
3. (a) Solve the ODE: $(2x+3y+7)dx+(3x-5y+2)dy=0$.

   (b) Find the orthogonal trajectory of the family of curves $x^2+y^2=r^2$, where r is the variable parameter.
4. Solve the ODE: $(x^2D^2+3xD+2)y=\cos(\log x)$, where $D=\frac{d}{dx}$.
5. Describe Kruskal and Prim's algorithm for finding the minimal spanning tree with examples.